

Esempio-1

Consideriamo il seguente problema di minimizzazione

$$f(x, y) = 4(x - 1)^2 + (y - 2)^2$$

soggetto ai seguenti vincoli

- $x + y \leq 2$;
- $x \geq -1$;
- $y \geq -1$;

Calcoliamo la Lagrangiana (moltiplico il II e III vincolo per -1)

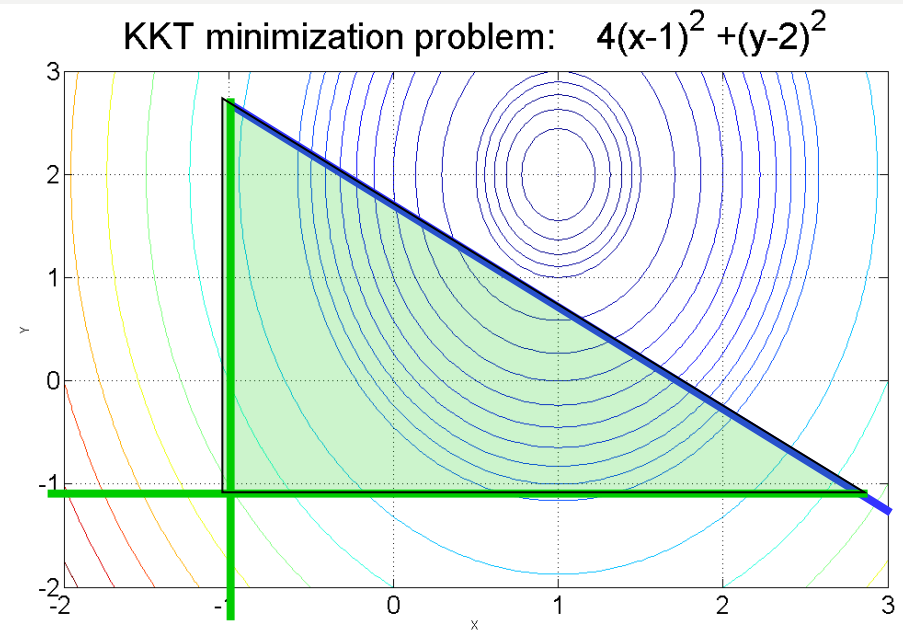
$$L(\underline{x}, \underline{\lambda}, \underline{\mu}) = f(\underline{x}) + \sum_{i=1}^m \lambda_i g_i(\underline{x}) + \sum_{j=1}^p \mu_j h_j(\underline{x})$$

$$L(\underline{x}, \underline{\lambda}, \underline{\mu}) = 4(x - 1)^2 + (y - 2)^2 + \mu_1(x + y - 2) - \mu_2(x + 1) - \mu_3(y + 1)$$

Vi sono tre vincoli di disuguaglianza, ognuno di essi può essere attivo oppure no, per un totale di 8 combinazioni possibili MA

→ tutti e tre i vincoli non possono essere attivi ($x+y=2$ & $x=-1$ & $y=-1$ non ha soluzione)

→ dall'intersezione di 2 vincoli si ottiene un solo punto



Esempio-2

Nel caso generale dobbiamo considerare tutte le combinazioni di vincoli attivi o non attivi

$$L(\underline{x}, \underline{\lambda}, \underline{\mu}) = 4(x-1)^2 + (y-2)^2 + \mu_1(x+y-2) - \mu_2(x+1) - \mu_3(y+1)$$

$$(1) \ x + y = 2 \Rightarrow L(x, y, \mu) = 4(x-1)^2 + (y-2)^2 + \mu_1(x+y-2)$$

$$(2) \ x = -1 \Rightarrow L(x, y, \mu) = 4(x-1)^2 + (y-2)^2 - \mu_2(x+1)$$

$$(3) \ y = -1 \Rightarrow L(x, y, \mu) = 4(x-1)^2 + (y-2)^2 - \mu_3(y+1)$$

$$(4) \ x + y = 2 \quad e \quad x = -1 \Rightarrow (x, y) = (-1, 3)$$

$$(5) \ x + y = 2 \quad e \quad y = -1 \Rightarrow (x, y) = (3, -1)$$

$$(6) \ x = -1 \quad e \quad y = -1 \Rightarrow (x, y) = (-1, -1)$$

$$(7) \ x + y = 2 \quad e \quad x = -1 \quad e \quad y = -1 \Rightarrow (x, y) = \emptyset$$

$$(8) \text{ Non vincolato: } L(x, y) = 4(x-1)^2 + (y-2)^2$$

Si ricorda che per le condizioni di complementarità ai vincoli non attivi corrispondono variabili lagrangiane uguale a zero

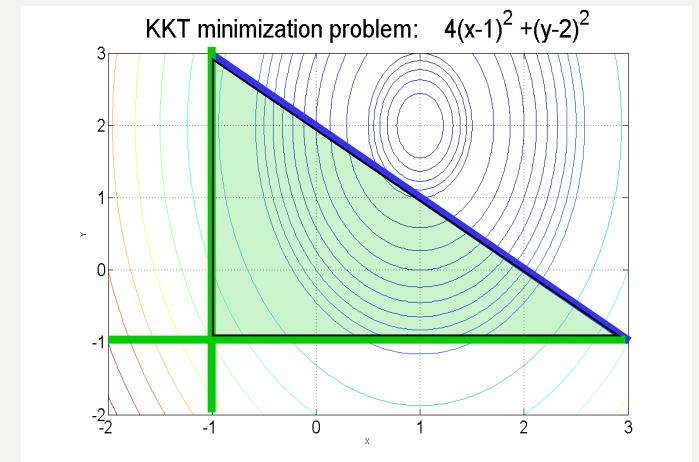


Esempio-3

Analizziamo ora caso per caso

$$(1) L(x, y, z) = 4(x - 1)^2 + (y - 2)^2 + \mu_1(x + y - 2)$$

$$\left. \begin{aligned} \frac{\partial L}{\partial \mu_1} &= x + y - 2 \\ \frac{\partial L}{\partial x} &= 8x - 8 + \mu_1 \\ \frac{\partial L}{\partial y} &= 2y - 4 + \mu_1 \end{aligned} \right\} \Rightarrow \left. \begin{aligned} x &= 2 - y \\ \mu_1 &= 8 - 8x \\ 2y - 4 + 8 - 8(2 - y) &= 0 \end{aligned} \right\} \Rightarrow \begin{aligned} (x, y) &= (0.8, 1.2) \\ f(x, y) &= 0.8 \\ \mu_1 &= 1.6 > 0 \\ \mu_2 &= \mu_3 = 0 \end{aligned}$$

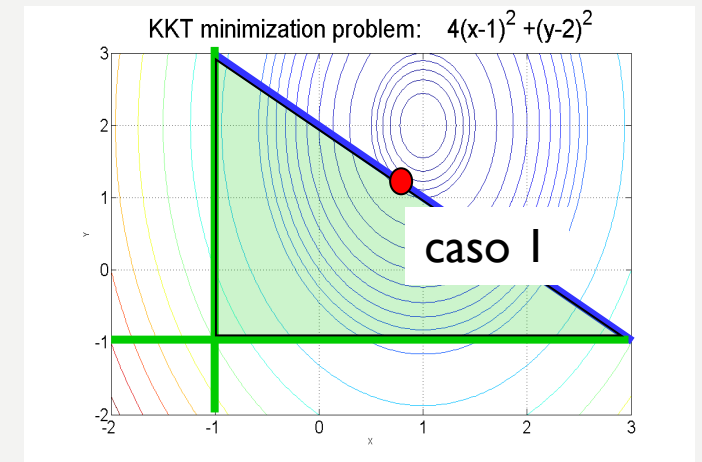


Esempio-3

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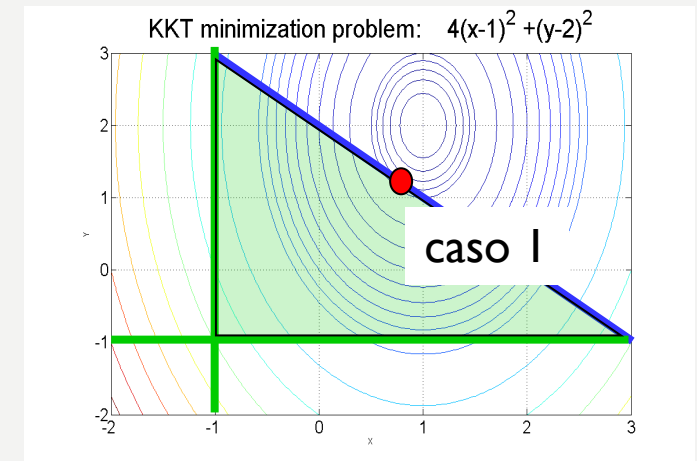
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$$(2) L(x, y, z) = 4(x - 1)^2 + (y - 2)^2 - \mu_2(x + 1)$$

$$\left. \begin{aligned} \frac{\partial L}{\partial \mu_2} &= -(x + 1) \\ \frac{\partial L}{\partial x} &= 8x - 8 - \mu_2 \\ \frac{\partial L}{\partial y} &= 2y - 4 \end{aligned} \right\} \Rightarrow \left. \begin{aligned} x &= -1 \\ \mu_2 &= -16 \\ y &= 2 \end{aligned} \right\} \Rightarrow \begin{aligned} (x, y) &= (-1, 2) \\ f(x, y) &= 16 \\ \mu_2 &= 16 < 0 \\ \mu_1 &= \mu_3 = 0 \end{aligned}$$



Esempio-3

Analizziamo ora caso per caso

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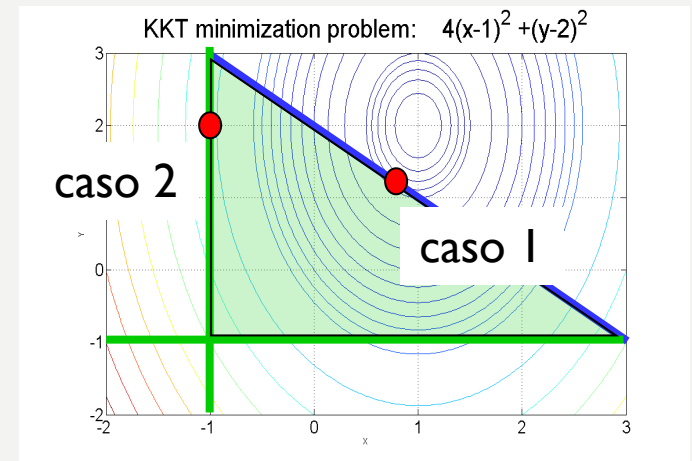
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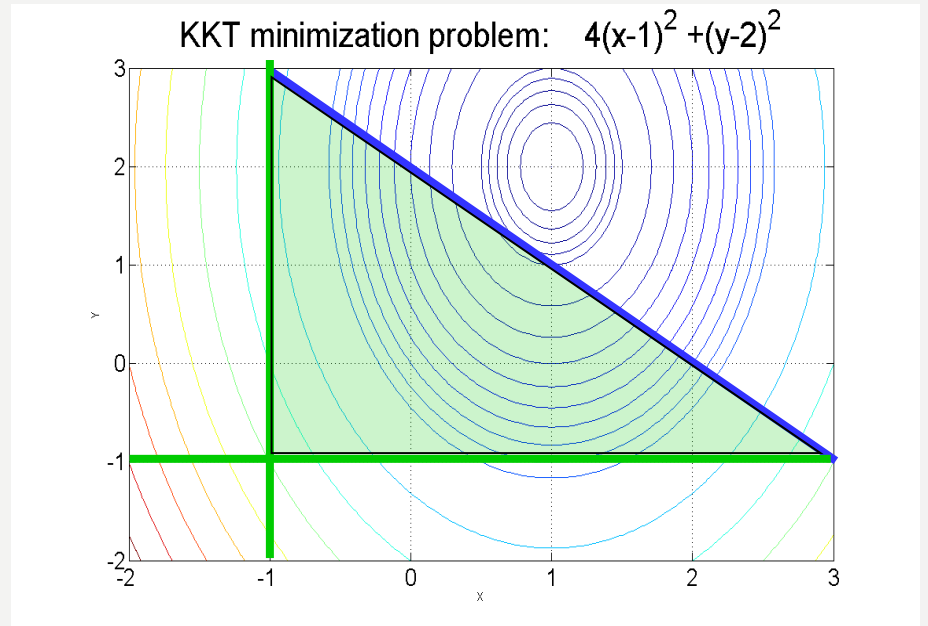
Moltiplicatore lagrangiano negativo !



Esempio-4

$$(3) \quad L(x, y, z) = 4(x - 1)^2 + (y - 2)^2 - \mu_3(y + 1)$$

$$\left. \begin{aligned} \frac{\partial L}{\partial \mu_3} &= -(y + 1) \\ \frac{\partial L}{\partial x} &= 8x - 8 \\ \frac{\partial L}{\partial y} &= 2y - 4 - \mu_3 \end{aligned} \right\} \Rightarrow \left. \begin{aligned} y &= -1 \\ x &= 1 \\ \mu_3 &= -6 \end{aligned} \right\} \Rightarrow \begin{aligned} (x, y) &= (1, -1) \\ f(x, y) &= 9 \\ \mu_3 &= -6 < 0 \\ \mu_1 &= \mu_2 = 0 \end{aligned}$$



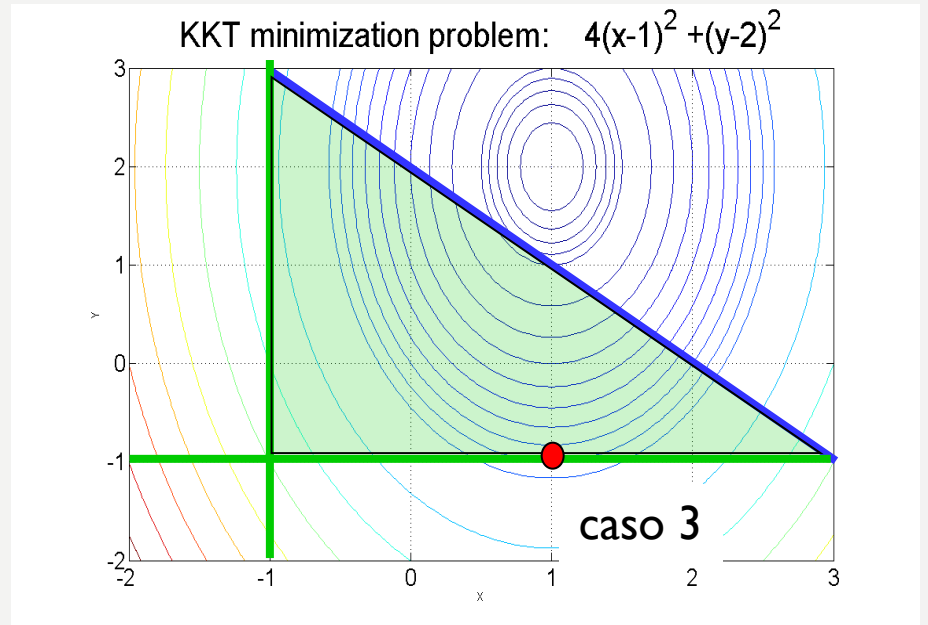
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Esempio-4

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Moltiplicatore lagrangiano negativo !



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Moltiplicatore lagrangiano negativo !

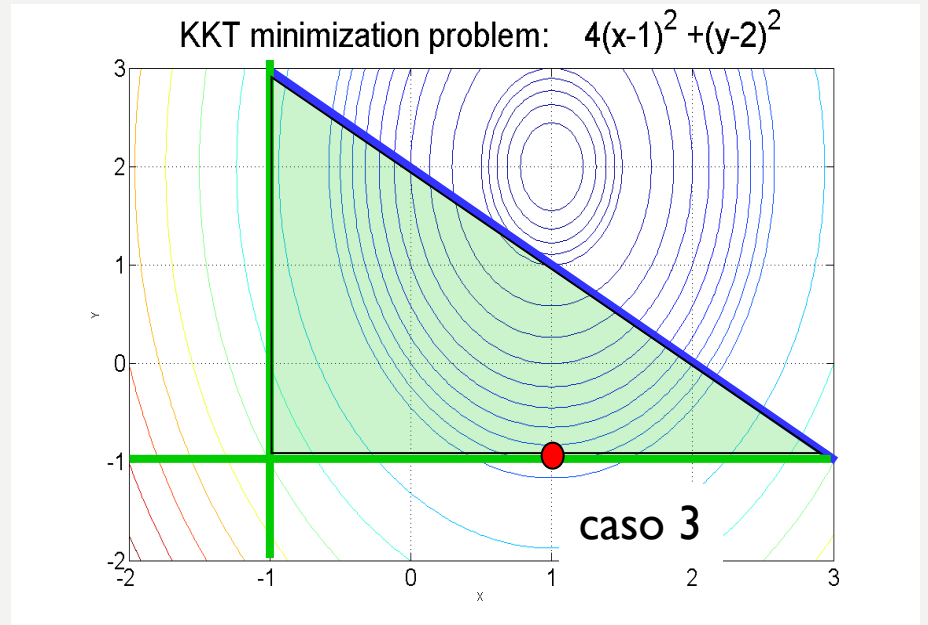
$$(4) \quad (x, y) = (-1, 3); \quad f(x, y) = 17$$

$$(5) \quad (x, y) = (3, -1); \quad f(x, y) = 25$$

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$$(8) \quad (x, y) = (1, 2); \quad f(x, y) = 0 \quad x + y = 3 > 2 \quad - \text{ non è nella regione ammissibile}$$



$$L(\underline{x}, \underline{\lambda}, \underline{\mu}) = 4(x - 1)^2 + (y - 2)^2 + \mu_1(x + y - 2) - \mu_2(x + 1) - \mu_3(y + 1)$$

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Moltiplicatore lagrangiano negativo !

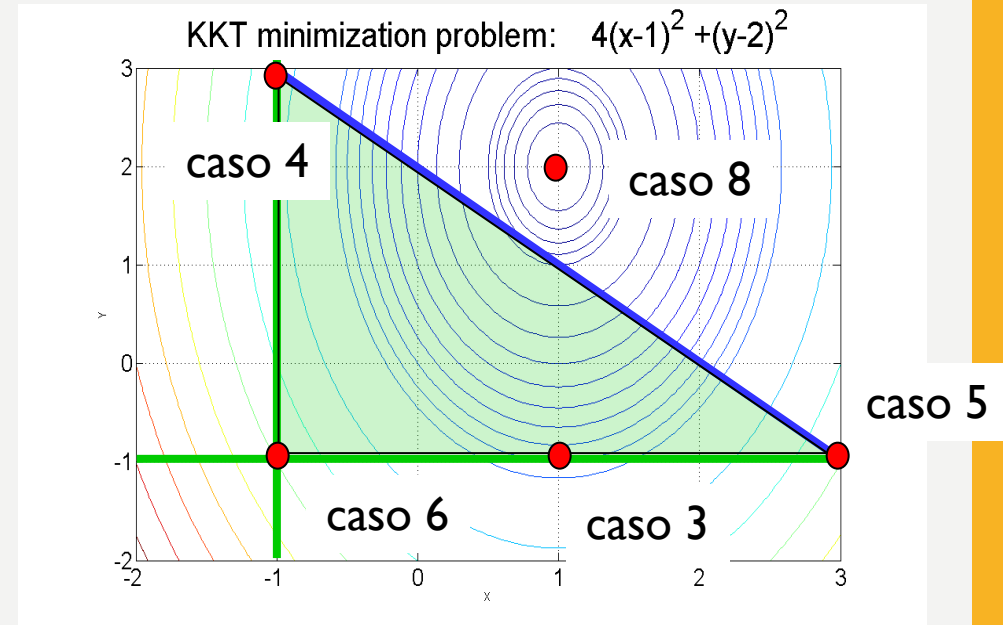
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$$L(\underline{x}, \underline{\lambda}, \underline{\mu}) = 4(x - 1)^2 + (y - 2)^2 + \mu_1(x + y - 2) - \mu_2(x + 1) - \mu_3(y + 1)$$

Esempio-5

Possiamo alla fine confrontare gli 8 casi che abbiamo considerato:

- (2) e (3) non soddisfano vincolo di non negatività del moltiplicatore lagrangiano
- (7) non ha soluzione
- (8) viola il vincolo $x+y \leq 2$.
- Tra i casi (1)-(4)-(5)-(6), il caso (1) ha il valore più basso di $f(x,y)$. $(x,y) = (0.8, 1.2)$; $f(x,y) = 0.8$

